

Table 1. Key findings summary

Sex-specific immune trajectory breakdown

Analysis	Result	Interpretation
PERMANOVA - Infection	$p = 0.032$	Infection alters immune profiles
PERMANOVA - Sex×HI×He	$p = 0.021$	Complex sex-specific hybrid effects
Dispersion - HI×Infection	$p = 0.00169$	Hybrid effects depend on infection
Male trajectory collapse	5.67 → 2.59 units	Males lose coordination when infected
Peak sex divergence	HI 0.2-0.4	Early hybrids most affected
Gene variance differences	0/19 genes	Systemic not gene-specific